

Incident Response Playbook – Windows Authentication Investigation

Playbook Information

Field	Value
Playbook Name	Windows Authentication Investigation
Incident Type	Suspicious Authentication Activity
Severity	Medium–High
Primary Data Sources	Windows Security Log, Wazuh Threat Hunting Dashboard
Related Event IDs	4624, 4625
MITRE ATT&CK	T1110 (Brute Force), T1078 (Valid Accounts), T1021.001 (Remote Desktop Protocol)

Trigger Conditions

Initiate this playbook when one or more of the following conditions are observed:

- Multiple failed logon attempts (Event ID 4625)
 - Successful logon following repeated failures
 - Unusual login time or location
 - Remote Desktop (Logon Type 10) authentication
 - Authentication alerts generated by Wazuh
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Phase 1 – Identification

Objective

Validate the authentication alert.

Actions

- Review Wazuh alert details.
- Verify Event ID (4624 or 4625).
- Identify affected user account.
- Record hostname and timestamp.
- Determine logon type.

Evidence

- Username
- Computer Name

- Event ID
 - Logon Type
 - Timestamp
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Phase 2 – Investigation

Objective

Determine whether the authentication is legitimate.

Actions

- Review authentication history.
- Check for repeated failed logins.
- Verify if the login occurred during normal business hours.
- Review source IP (if available).
- Correlate with PowerShell, Sysmon, or network events.
- Determine whether the account normally performs remote administration.

Questions

- Is the user authorized?
 - Was the login expected?
 - Were additional suspicious events generated after authentication?
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Phase 3 – Containment

If malicious activity is confirmed:

- Isolate the endpoint.
 - Disable or lock the compromised account.
 - Block malicious IP addresses (if identified).
 - Preserve logs for forensic analysis.
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Phase 4 – Eradication

Remove attacker artifacts.

- Remove malicious scheduled tasks.
- Delete persistence mechanisms.
- Scan for malware.
- Reset affected credentials.

- Remove unauthorized user accounts (if created).

Phase 5 – Recovery

Restore normal operations.

- Reconnect the endpoint.
- Verify authentication services.
- Monitor for recurring login attempts.
- Confirm no additional suspicious alerts are generated.

Phase 6 – Lessons Learned

Document the investigation.

Include:

- Root cause
- Timeline of events
- Indicators of Compromise (IOCs)
- Response actions taken
- Detection improvements
- Recommendations for preventing recurrence

Investigation Workflow

Authentication Alert

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Validate Alert

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Collect Authentication Evidence

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Correlate Related Events

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Assess Severity



Contain Threat



Eradicate



Recover



Lessons Learned

Analyst Checklist

- ☐ Validate the alert.
- ☐ Verify Event IDs (4624/4625).
- ☐ Identify the affected account.
- ☐ Determine the logon type.
- ☐ Review login history.
- ☐ Correlate with endpoint activity.
- ☐ Assess incident severity.
- ☐ Contain the threat if malicious.
- ☐ Document findings.
- ☐ Recommend detection improvements.

Expected Outcome

By following this playbook, the SOC analyst can:

- Validate authentication-related alerts.
- Distinguish legitimate logins from suspicious activity.
- Correlate authentication events with other endpoint telemetry.
- Respond consistently using a structured Incident Response process.

- Document findings for future investigations and continuous improvement.
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